Requirements:

1. Implement three different trading strategies:
    - Moving Average Crossover (as covered in the chapter)
    - Breakout Strategy (buy when price breaks above recent high, sell when it breaks below recent low)
    - RSI Mean Reversion (buy when RSI is oversold, sell when RSI is overbought)
2. Create a unified backtesting framework that:
    - Applies all three strategies to the same set of stocks (at least 5 diverse stocks)
    - Uses the same historical period for all tests (minimum 3 years)
    - Ensures fair comparison by applying the same capital constraints and transaction costs
    - Generates consistent performance metrics across all strategies
3. Evaluate each strategy individually with detailed metrics:
    - Risk-adjusted returns (Sharpe and Sortino ratios)
    - Drawdown analysis (maximum drawdown, average drawdown, recovery time)
    - Win/loss statistics (win rate, average win, average loss, largest win, largest loss)
    - Market exposure (percentage of time invested vs. in cash)
4. Analyze how each strategy performs in different market conditions:
    - Bull markets vs. bear markets
    - High volatility vs. low volatility periods
    - Different sectors or asset classes